

COMPUTER VISION ASSIGNMENT 2

21 AUGUST 2026

Q1. Line Structure Detection using LoG and Hough Transform

Select an image containing prominent linear structures. First, convert the image to grayscale and detect edges using the **Laplacian of Gaussian (LoG)** operator. Obtain the edge map using zero-crossing detection and repeat the process using a suitable threshold to suppress weak responses.

Using the resulting edge image, apply the **Hough Transform** for line detection. Visualize the detected lines overlaid on the original image.

Compare the results obtained using the two LoG threshold settings and discuss how the choice of threshold affects the number of detected lines and false detections. *You may use built-in functions for the Hough Transform.*



Figure 1: Line Structure Detection

Q2. Robust Circle Detection using Hough Transform

Consider an image containing multiple circular objects with varying sizes, partial boundaries, and/or noise. Apply the **Hough Transform for circle detection** to identify the circular objects.

Display the detected circles on the original image and investigate how changing the **radius range** and **detection threshold** affects the results. Report the parameter settings that provide the most accurate detection and briefly explain the presence of any false or missed detections.



Figure 2: Circle Detection

Q3. The Role of Noise in Image Thresholding

Background: Image noise heavily impacts the quality of thresholding by broadening and overlapping intensity histogram peaks, making it difficult to select a clean boundary between object and background.

1. Take a clean, noiseless grayscale image and simulate image degradation by adding Gaussian noise with a mean of 0 and varying standard deviations (e.g., $\sigma = 10$ and $\sigma = 50$).
2. Display the noisy spatial-domain images alongside their corresponding intensity histograms.
3. Apply a global threshold to the noisy images and discuss how increasing noise levels affect the histogram separation and the resulting binarized segmentation output.

Q4. Global Thresholding using Otsu's Method

Background: Otsu's method is an automatic, variance-based global thresholding technique used to separate an image into foreground and background classes. It works by searching through all possible threshold values to find the one that minimizes the intra-class (within-class) variance, which is mathematically equivalent to maximizing the inter-class (between-class) variance. For a comprehensive overview, you can refer to the [Guide on Otsu's Method](#).

Select a grayscale image with a clear bi-modal intensity histogram (e.g., an object against a contrasting background). Implement Otsu's thresholding method from scratch to automatically determine the optimal global threshold value T .

1. Compute the normalized histogram of the image to obtain the probability distribution of intensity levels.
2. Calculate the between-class variance $\sigma_B^2(T)$ for all possible threshold values T .
3. Plot the between-class variance as a function of T and identify the optimal threshold that maximizes $\sigma_B^2(T)$.
4. Apply the optimal threshold to binarize the image and display the resulting foreground-background segmentation.
5. Discuss how Otsu's method performs when the image histogram is unimodal or has significant illumination variations.